

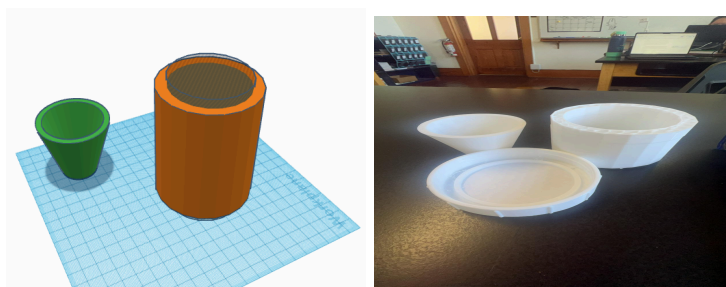
Fruit Fly Trap

Team: Cheesecake

Element G: Construction of a Testable Prototype

Our boarding school residents need a way to remove fruit flies from their dormitories. This is because residents are irritated by the fruit flies and they could also pose a health risk. The fruit flies are attracted to a food item that was left in the dormitories a few weeks ago. Fruit flies have a short reproduction time so the quicker they are removed the easier it will be because there are less flies to remove. Quality of life in the dormitories would also improve if the fruit flies were to be removed.

The first prototype consisted of having a funnel screw to the top of the bottle so there is little room for the fly to go into. We thought about it using examples we saw and we decided to try something new. The prototype was going to be 3D printed so it is efficient and we can add new parts later on. The fruit fly trap is going to be able to keep the flies from leaving after they enter and they will drop into the bottom. You can unscrew the cap on the bottom to empty out the flies at the end.



We changed the prototype the second time and added little divots to the side. The divots were to make sure the cap on the bottom could screw on. We added a piece for the funnel to fit inside so it can be removed easily. We added the funnel to the top but we changed the size of it

so that it could fit into the top.

The final design consisted of having the funnel and the cap on it and we also fixed some small things on our prototype so that it could be in our design requirements. We wanted our design to fit everyone's problem and that is a reason why we ended up changing some of the parts to the fly trap. The fly trap was originally going to be built using different materials but we wanted our design to be stable so that is why we 3D printed the prototype.

Our final prototype was also 3D printed and it was the same as the other prototypes but with changes. The prototype will be able to test the design requirements because we will be testing for things like weight and stability. Some of the tests are going to be focusing on if the design is going to be able to function in different areas or if the prototype will need to be changed again to match the requirements.

The prototype is going to be able to test if the trap can get rid of 80% of the flies in the room. We will be able to test that by placing the design in the room and wait a couple of days to see if the fruit fly population has gone down in the room. We also have to test for odor and to do that we are going to see if we can smell the fruit fly trap. We will do this by placing the liquid inside of the trap and waiting a while to see if the room smells like the trap or a certain area does.

Some parts of our design can not be tested and some can. Some of the things that can be tested are high priority and some of them a low priority. Some of the tests like if the fruit fly trap will react to different temperatures we can not test and that is because it will not be a main concern for the person buying it. We also can not test if the trap will have to be maintained a lot and that is because when the trap is placed in the room the customer is not going to want to have to work on it all the time so we can not test that. We can test things like the stability and durability and those tests will be what the customer is going to want to have as a trap.